

Fraud Prediction using IBM watsonx.ai



Preface

Overview

Automation and artificial intelligence (AI) are transforming businesses and will contribute to economic growth via contributions to productivity. They will also help address challenges in areas of healthcare, technology & other areas. At the same time, these technologies will transform the nature of work and the workplace itself. In this code pattern, we will focus on building state of the art systems for churning out predictions which can be used in different scenarios. We will try to predict fraudulent transactions which we know can reduce monetary loss and risk mitigation. The same approach can be used for predicting customer churn, demand and supply forecast and others. Building predictive models require time, effort and good knowledge of algorithms to create effective systems which can predict the outcome accurately. With that being said, IBM has introduced AutoAI which will automate all the tasks involved in building predictive models for different requirements. We will get to see how AutoAI can churn out great models quickly which will save time and effort and aid in faster decision-making process.

Industry Use-case

A. Fraud detection in the insurance business

Headquartered in Aurora, just outside of Chicago, Northeast Insurance Company, (NIC) employs over 5000 people across the continental United States.

During its fifty-year history, NIC has been struggling to detecting potentially fraudulent activity and has turned to IBM for their data science and AI offerings to predict fraud with insurance claims, before the claim is settled.

B. Business challenge story

The global Fraud Detection and Prevention (FDP) market size is expected to grow from USD 20 billion to 63.5 billion by 2023, according to various analyst reports (i.e. "[Fraud Detection and Prevention Market by solution](#)"). Predictive analytics segment is projected to be the largest contributor to the FDP market during the forecast period.

Predictive analytics solutions help enterprises identify the possibilities of fraud incidents by analyzing the current data. The solutions are used to identify potential threats, payment frauds, frauds in insurance processes, and credit/debit card frauds. Organizations are trying to impart these solutions for predicting fraud or suspicious activity and their pattern to help drastically reduce losses due to frauds.

Fraud analytics solutions employ sophisticated analytics and predictive modeling to identify potential fraud in real time during data entry, rather than during a later batch run after a transaction is complete. It can be applied to claims and underwriting fraud. Regional Tier 2 and 3 insurers are more likely to adopt SaaS-based point solutions for fraud analytics use cases. Larger insurers are implementing these solutions via professional services providers and system integrators.

Drivers usually sign a six-month policy with an auto insurance policy. Each month, or all at once, the driver pays a fee, or **premium**, to the company. There are a few things that determine the cost of the policy: the type of car insured (particularly its safety record and how expensive it is to repair) the driver's record (the more speeding tickets the driver has incurred, the riskier he is) and even age (teenagers cost more to insure because they're less experienced drivers, and therefore a bigger risk.) Lower-cost premiums are enjoyed by drivers with fewer accidents and tickets on their records, part-time drivers, people who take driver education courses, and families with multiple cars.



PAIN POINTS

Information siloed, overload, difficult to see clearly

Using AI/ML for fraud detection is not new. However, typical organization contains multiple fraud departments, each with its own internal point-solution which monitors fraud for that specific channel, product, or fraud type. Structured and unstructured data collected internally and externally but very few of these point-solutions share data. Each uses varying analytical techniques across channels and transaction systems, which results in not having a complete view of risk exposures across the institution. Cannot see patterns or behaviors that would spark a concern that fraudulent activity is crossing multi-business lines because the observation space is too narrow.

Difficult to predict fraud

Rare occurrences create an imbalance in the classification of fraud detection models and makes detection challenging.

Shift to increased digital and mobile customer platforms led to transactions being executed more quickly, leaving banks and processors with less time to identify, counteract, and recover the underlying funds. As quickly as new technology is used to identify fraudsters, they themselves are identifying new ways of defrauding the bank. For instance, identity theft is mutating from card skimming to account takeovers (ATO). Synthetic identity, a scenario where fraudsters combine fragments of stolen or fake information to create a new identity and apply for financial products.

Cost of (near) real-time detection is high

Organizations need to identify anomalies accurately and efficiently at the level of accounts, merchants, cardholders and locations.

False positives require manual investigations through providing content analytics across primary internal and external data sources

Fraud detection – meaning detecting fraudulent behavior after it occurs – forcing companies to set aside money and resources for the inevitable losses they will incur, costing financial institutions millions of dollars and destroying the customer experience. Financial institutions need to get in front of the problem and focus on fraud prevention.

72%

cite fraud as a growing concern over the past 12 months and nearly **63%** report the same or higher levels of fraudulent losses over that same period

\$44B

Worldwide losses due to fraud by 2025

25%

of declined sales transactions for e-commerce merchants were false positives.

Tools

- IBM Watson Studio: Analyze data using RStudio, Jupyter, and Python in a configured, collaborative environment that includes IBM value-adds, such as managed Spark.
- IBM Auto AI: The AutoAI graphical tool in Watson Studio automatically analyzes your data and generates candidate model pipelines customized for your predictive modeling problem.
- IBM Cloud Object Storage: An IBM Cloud service that provides an unstructured cloud data store to build and deliver cost effective apps and services with high reliability and fast speed to market. This code pattern uses Cloud Object Storage.

Understanding the Data

Let's begin by accessing the CSV file required for the following steps.

1. Download the CSV file: https://github.com/apischdo/AI_Periodic_Table/blob/main/AutoInsClaims.csv

It is recommended that you use Firefox, IE or Mozilla (Edge and Safari are not recommended). Once you access the Github page, click **Raw**. Then, from the ensuing page, right-click and save the file as CSV (change file type to **All file types**).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	HOUSEHOLD	DRIVER_ID	POLICY_ID	CLAIM_ID	INCIDENT_CD	DESCRIPTION	CLAIM_STAT	ODOMETER	LOSS_EVENT	CLAIM_INIT	POLICE_REPT	CLAIMS_AT	LOSS_LOCAT	LOSS_LOCAT	CLAIM_AMO	FLAG_FOR	I_PRIMARY	DI_START_DATE	EXPIRY_DATE	MODEL_YEA	MAKE
2	CH42335	XZJ2837	NW5567882	A-2017-UU9	3			157654.9	4/25/17	4/28/17			41.9021031	-87.755624	35765	1	XZJ2837	9/19/16	9/19/17	2010	Nissan
3	IH49805	VVR6423	UR4864804	A-2018-FI48	3			226154.5	8/26/18	8/31/18	0	1	41.9635619	-87.731397	1909	0	VVR6423	6/14/17	6/14/18	2008	Dodge
4	AF28736	UQM2512	RR8595908	A-2016-ZG6	1			83968.6	1/7/16	1/11/16	0	2	41.7366016	-87.604968	25730	1	UQM2512	8/24/15	8/23/16	2009	Chevrolet
5	EF53594	YDT5591	RN5640634	A-2016-NG7	1			309570.3	12/11/16	12/18/16	0	1	41.9099253	-87.731557	40880	1	YDT5591	7/25/15	7/24/16	2002	Chevrolet
6	LD32277	ONM5465	YY1229530	A-2017-ZO8	3			136633.9	6/6/17	6/8/17	0	1	41.9237502	-87.789881	2130	0	IJZ5688	1/8/16	1/7/17	2010	Honda
7	DM94074	GBU7751	XP3473763	A-2018-XB4	1			326514.1	3/6/18	3/19/18	0	1	41.9092571	-87.785057	1970	0	BUK6977	1/15/17	1/15/18	2004	Ford
8	MD38210	CBR4335	US5444269	A-2017-XP75	2			58477.9	8/13/17	8/18/17	0	1	41.9283951	-87.796468	2290	0	CBR4335	7/8/16	7/8/17	2012	Chevrolet
9	GL77908	HZF3884	XR1994270	A-2017-QY9	1			176476.9	2/10/17	2/18/17	0	1	41.8857155	-87.728376	2990	0	QAX1481	8/11/16	8/11/17	2010	Kia
10	BA26199	CE9523	VP6368585	A-2018-LB81	3			277812.7	2/16/18	2/24/18	0	1	41.895054	-87.745662	1170	0	CE9523	10/13/17	10/13/18	2006	Nissan
11	EA38976	HF7408	YP9758006	A-2016-QD3	5			190541.3	12/28/16	1/11/17	0	1	41.8919756	-87.61458	2090	0	HF7408	6/23/15	6/22/16	2008	Kia
12	GB64343	PXE3728	XY6800348	A-2017-CK71	1			290975.1	8/12/17	8/25/17	0	1	41.9112234	-87.638656	2130	0	PXE3728	4/14/16	4/14/17	2005	Kia
13	JG99629	OKH5337	ZK6994471	A-2018-WF1	3			159873.2	7/30/18	7/30/18	1	5	41.7358355	-87.667866	33040	1	OKH5337	2/22/17	2/22/18	2010	Hyundai
14	HH1231	RQZ1566	ZR4462879	A-2018-VT47	1			391866.9	2/22/18	2/26/18	0	1	41.7341643	-87.551211	2400	0	FSC3949	1/10/17	1/10/18	1998	Dodge
15	KI98597	VRX5780	ZW4263453	A-2017-SA27	3			384000.6	2/6/17	2/6/17	0	3	41.7918521	-87.801378	21330	1	VRX5780	6/4/16	6/4/17	2000	Toyota
16	CG61685	TIA1702	UW8176531	A-2018-EJ36	2			251644.8	7/10/18	7/18/18	1	1	42.0059308	-87.68066	3427	0	TIA1702	4/10/17	4/10/18	2007	Nissan
17	AB12181	OB08151	RW8652538	A-2017-CT58	2			300279.3	7/29/17	8/10/17	0	1	41.6934005	-87.612322	35385.5	1	OB08151	11/25/16	11/25/17	2004	Chevrolet
18	BI63668	WDB2749	PQ3189850	A-2017-CJ81	2			73419.5	12/30/17	1/3/18	0	1	41.8401446	-87.661339	1570	0	OZP7842	4/6/16	4/6/17	2010	Honda
19	MG83193	XIB5149	KN9015815	A-2018-LF97	2			118923.9	1/16/18	1/28/18	0	1	41.9751749	-87.768344	2370	0	XIB5149	11/21/17	11/21/18	2007	Jeep
20	UI01709	CRP8660	SV1659058	A-2017-ML6	1			182681.3	4/25/17	5/7/17	1	1	41.8901726	-87.649992	26250	1	CRP8660	7/23/17	7/23/17	2009	Chevrolet
21	BC94182	PEQ6155	UO8163328	A-2018-WZ4	5			160429.9	2/10/18	2/19/18	1	3	41.8406579	-87.725899	25830	1	PEQ6155	7/11/17	7/11/18	2011	Nissan
22	MF96553	PPH8400	OT0287150	A-2018-DH45	2			165363.7	1/23/18	1/24/18	1	4	41.8847051	-87.667046	23483	1	PPH8400	4/23/17	4/23/18	2010	Hyundai
23	BC90853	JOA9170	TW2172802	A-2016-RW6	5			240639.2	12/1/16	12/10/16	1	1	41.7361826	-87.64366	2173.5	0	JOA9170	8/20/15	8/19/16	2005	Chevrolet
24	AK42388	KNQ4268	PP7360289	A-2017-KO8	4			388591.3	7/12/17	7/12/17	1	1	41.7620304	-87.611696	1750	0	KNQ4268	4/17/16	4/17/17	2001	Dodge
25	HB16909	SRU2950	UY6516260	A-2016-CA74	1			59498.7	1/6/16	1/8/16	1	1	42.0064221	-87.67812	20310	1	SRU2950	4/18/15	4/17/16	2013	Kia
26	FA97534	EHL9301	OS6708536	A-2018-PV57	1			94043.6	10/13/18	10/25/18	0	1	41.8136018	-87.704338	2810	0	BIN1474	10/1/17	10/1/18	2014	Chevrolet
27	CF57572	CY6373	QV4842191	A-2017-JI571	4			69305.4	7/19/17	7/19/17	1	1	41.9969237	-87.806945	2370	0	WKB8440	3/6/16	3/6/17	2014	Mazda
28	ME37855	NCL2868	VT9807337	A-2018-CK21	5			207764.4	11/14/18	11/13/18	0	1	41.9522275	-87.727425	1320	0	NCL2868	1/14/17	1/14/18	2008	Chevrolet
29	LA79549	YWA1319	PQ5439335	A-2017-FI21	5			302710	5/31/17	6/12/17	0	5	41.9768951	-87.69228	30840	1	YWA1319	4/4/16	4/4/17	2001	Acura
30	II25056	CVI8835	OJ5466096	A-2017-YI13	4			109080	3/1/17	3/13/17	0	3	41.8807808	-87.730232	1230.5	0	CVI8835	5/20/16	5/20/17	2012	Dodge
31	CK11297	HIR9014	RR5447110	A-2018-QW1	4			69640.2	9/5/18	9/11/18	0	1	41.8927529	-87.760709	1020	0	HIR9014	7/11/17	7/11/18	2015	Toyota
32	KJ99727	MDE6706	RQ6230412	A-2018-AC52	2			281332.9	12/14/18	12/19/18	1	1	41.9383457	-87.712307	2470	0	MDE6706	5/17/17	5/17/18	2004	Honda
33	GF42714	NK28115	ST5381628	A-2016-RA2	1			176864.1	7/14/16	7/25/16	0	2	41.9064302	-87.648306	31020	1	NK28115	8/23/15	8/22/16	2008	Hyundai
34	BD67160	SAD8006	YS2739165	A-2018-CE38	3			223253.6	11/26/18	11/30/18	0	1	41.890782	-87.631252	1890	0	SAD8006	2/8/17	2/8/18	2009	Toyota
35	MJ57549	OGG2359	TOS690517	A-2018-CX48	3			139853.9	7/20/18	7/20/18	0	1	41.8924442	-87.6372	1932	0	OGG2359	8/3/17	8/3/18	2012	Hyundai
36	FI15095	MW43612	NO2626234	A-2016-VA7	1			210844.5	9/24/16	9/26/16	0	1	41.8181867	-87.743319	1730	0	MW43612	7/25/15	7/24/16	2005	Kia
37	ID61421	EXX9506	YOS050770	A-2016-EL66	2			282275.4	8/6/16	8/20/16	0	3	41.8824105	-87.637211	2170	0	EXX9506	9/15/15	9/14/16	2004	Ford
38	MA52163	LSI8871	OR4862688	A-2017-TP82	1			224281.5	12/27/17	12/29/17	1	2	41.931362	-87.723151	20380	1	LOD2085	6/27/16	6/27/17	2007	Dodge

Open the file with Excel and notice that the columns are in string format and some clearly need to be numeric and that is just at a first glance. Let's take a moment and understand the significance or more aptly,

the predictability of each of the columns. Clearly, you can tell some of that info will not help you in predicting fraud. We need to remove the noise from the signal.

Take a moment, and note the column names: which may prove relevant? which are simply not needed? What if you used the date columns to calculate lapsed days from time of accident until reporting it? Or noted if the claims are being filled too close to the policy expiration date? Was there a police report? Let's look at expired licenses at the time of submitting the claim. What about clients with low mileage discounts (7500 per year) that do not have low mileage?

Take a moment and consider the table below bearing in mind the questions asked above, let's discuss that in a group if feasible, because before long, you will be immersed in feature engineering activities.

As you may have noted, there is a column named: **FLAG_FOR_FRAUD_INV**. Think of this column as training data for a supervised machine learning system. The rows with a value of 1 (True) have been verified and classified as fraud. That is known, not a prediction. We are going to build a model using this "training set" for data to predict future behavioral patterns that may then be flagged as potential fraud on a new CSV file never seen before by Watson Studio.

Feature name	R for remove P for predictable and why do you think that?
HOUSEHOLD_ID	R
DRIVER_ID	R
POLICY_ID	R
CLAIM_ID	R
INCIDENT_CAUSE 1 = driver error 2 = natural causes 3 = other driver error 4 = crime 5 = other causes	K
DESCRIPTION	R
CLAIM_STATUS 1 = open 2 = approved 3 = paid 4 = flagged for fraud 5 = denied 6 = appeal	R
ODOMETER_AT_LOSS	K
LOSS_EVENT_TIME	K
CLAIM_INIT_TIME	K
POLICE_REPORT 1 = there was police report 0 = no police report	K
CLAIMS_AT_LOSS_DATE (# of claims per individual)	K
LOSS_LOCATION_LAT	K
LOSS_LOCATION_LONG	K
CLAIM_AMOUNT	K
FLAG_FOR_FRAUD_INV	K (THIS IS YOUR X-AXIS)
PRIMARY_DRIVER_ID	R
START_DATE	K
EXPIRY_DATE	K
MODEL_YEAR	R

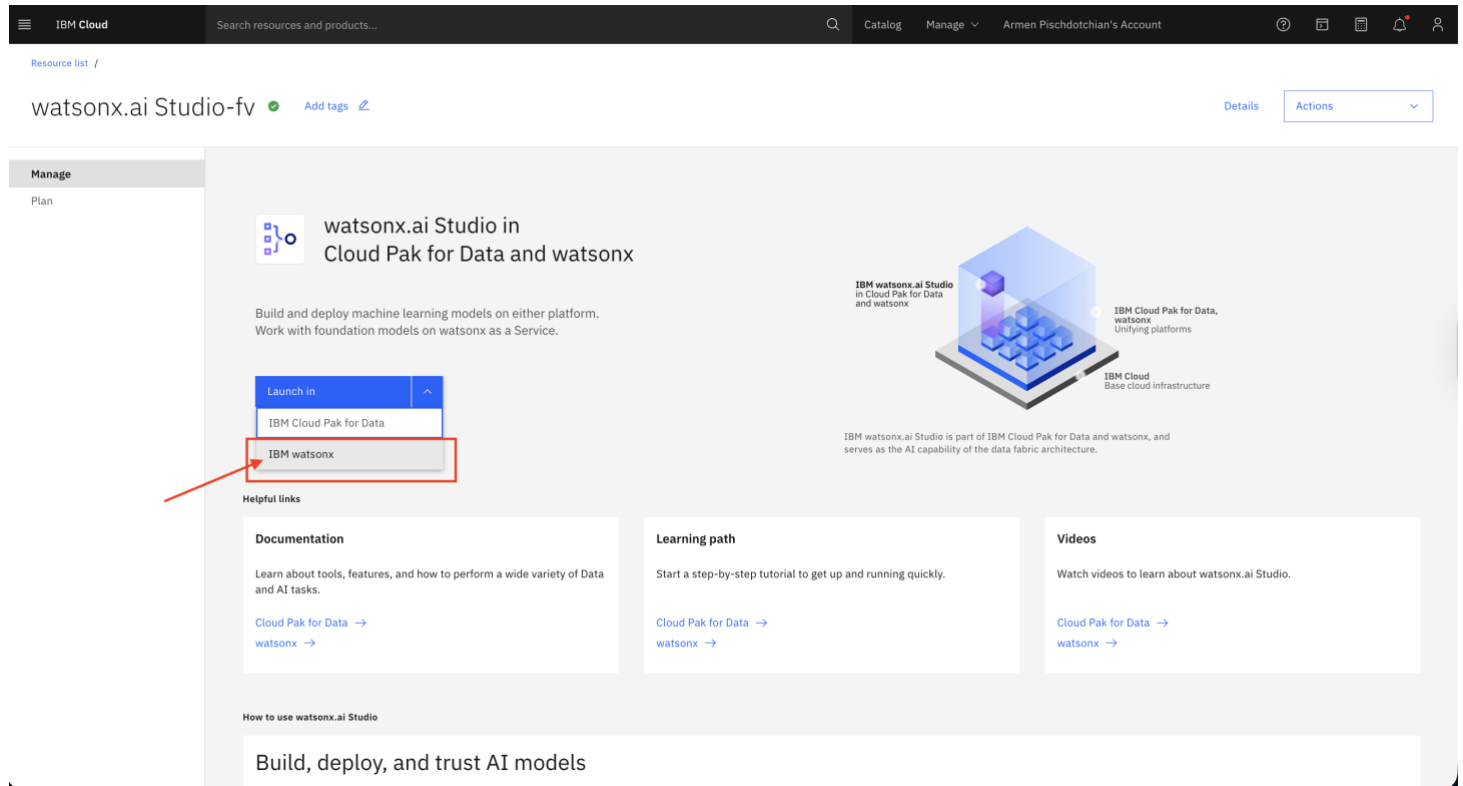
MAKE	R
MODEL	R
PLATE	R
COLOR	YOUR CHOICE
INITIAL_ODOMETER	K
LOW_MILEAGE_USE	K
FIRST_NAME	R
LAST_NAME	R
GENDER	K
BIRTHDATE	K
SSN	R
DRIVERS_LICENSE_ID	R
DRIVERS_LICENSE_EXPIRY	K
DRIVERS_LICENSE_STATE	WHAT DO YOU THINK?
DATE_AT_CURRENT_ADDRESS	K
CONTACT_NUMBER	R
EMAIL	R
COMMUTE_DISCOUNT	K

You are now ready for the detailed steps.

Build an AI model using AutoAI

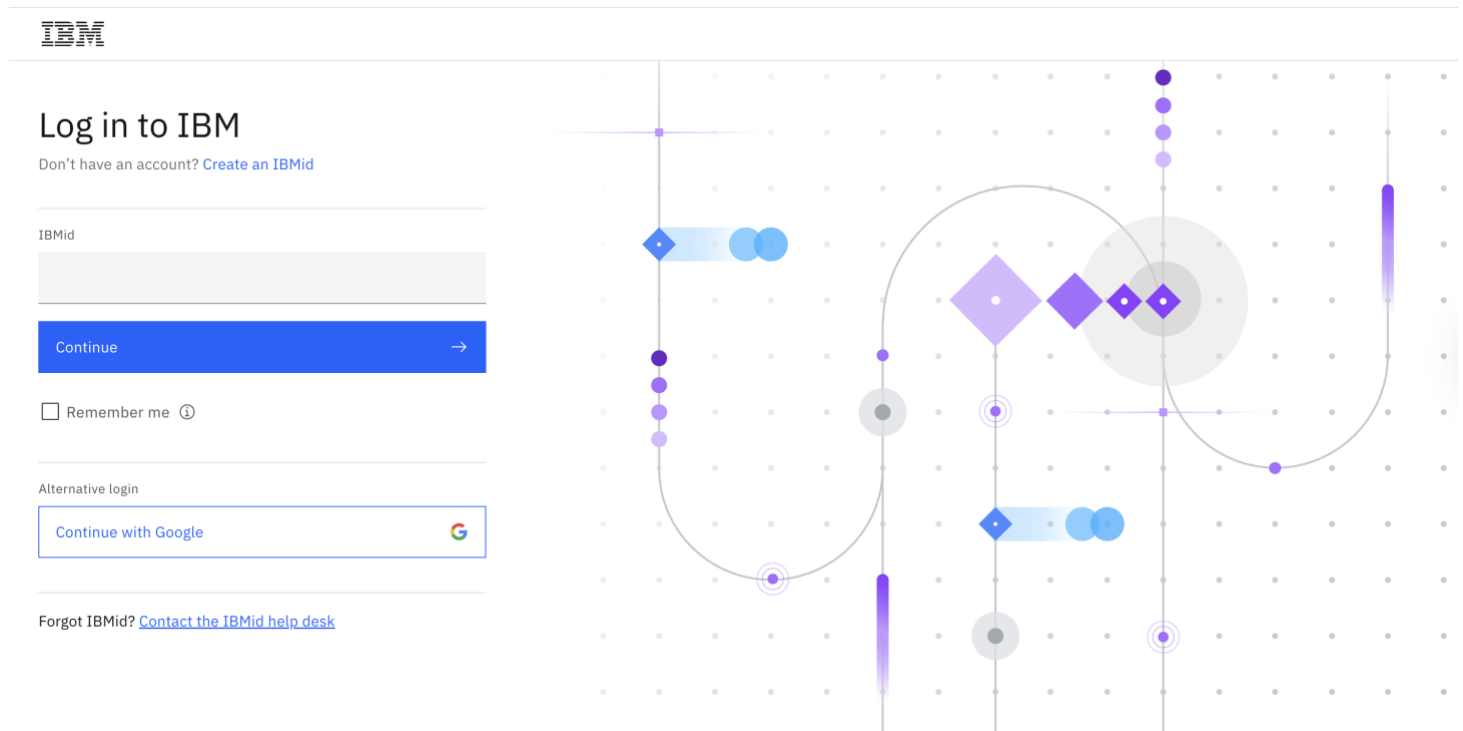
Create a new Watson Studio project

2. Provision **watsonx.ai Studio** (search for studio and select the first option).
3. Click **Launch in watsonx** (second option from the drop-down list)



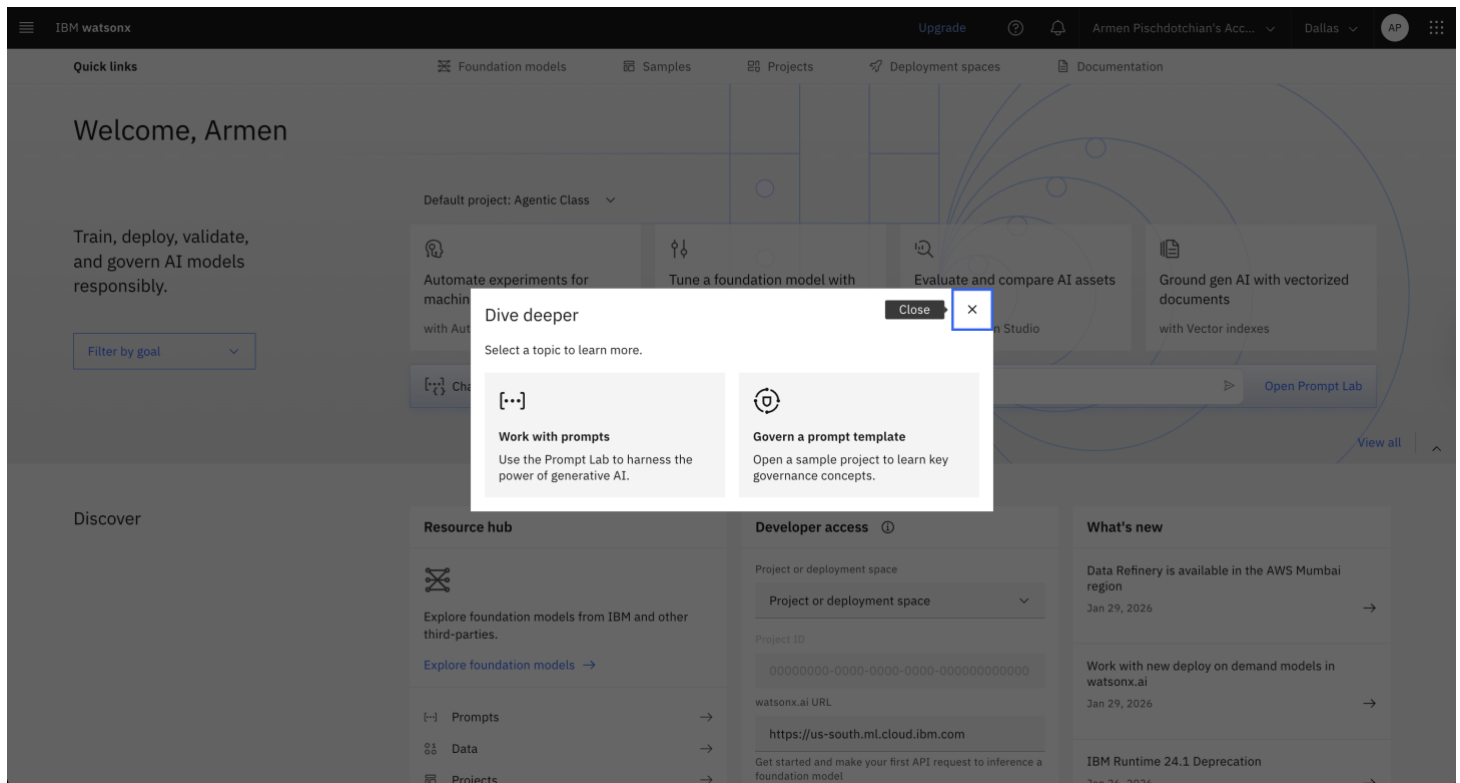
The screenshot shows the IBM Cloud console interface. At the top, there's a search bar and navigation links. Below, the 'Resource list' shows 'watsonx.ai Studio-fv'. The 'Manage' tab is selected, showing a 'Plan' section. The main content area displays 'watsonx.ai Studio in Cloud Pak for Data and watsonx'. It includes a description: 'Build and deploy machine learning models on either platform. Work with foundation models on watsonx as a Service.' Below this, a 'Launch in' dropdown menu is open, showing 'IBM Cloud Pak for Data' and 'IBM watsonx'. A red arrow points to 'IBM watsonx'. To the right, there's a diagram showing the architecture: 'IBM watsonx.ai Studio in Cloud Pak for Data and watsonx' on top of 'IBM Cloud Pak for Data, watsonx. Unifying platforms', which sits on 'IBM Cloud Base cloud infrastructure'. Below the diagram, text states: 'IBM watsonx.ai Studio is part of IBM Cloud Pak for Data and watsonx, and serves as the AI capability of the data fabric architecture.' Further down, there are sections for 'Helpful links', 'Documentation', 'Learning path', and 'Videos', each with links to 'Cloud Pak for Data' and 'watsonx'.

4. Login into Watson Studio.



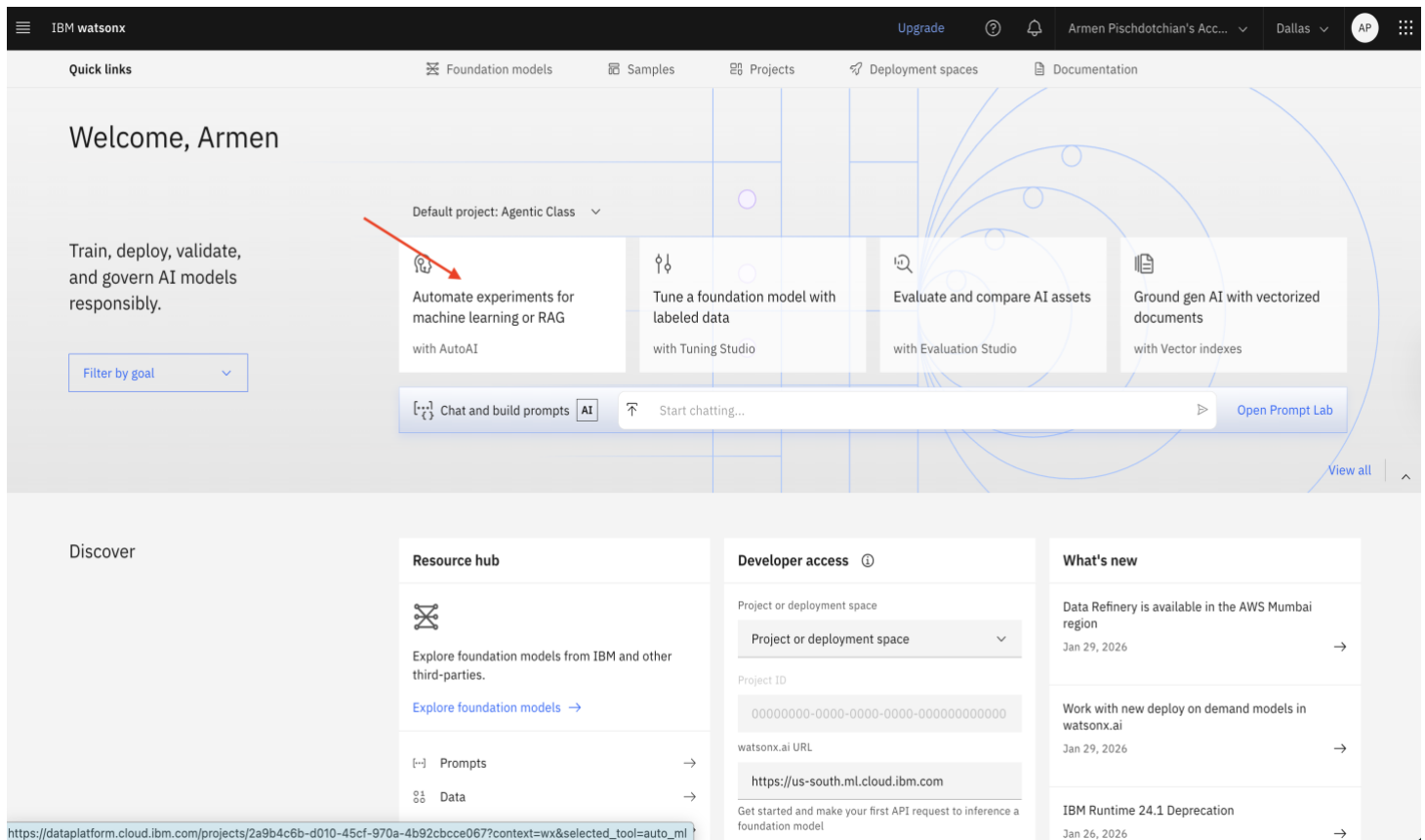
The screenshot shows the IBM login page. At the top left is the IBM logo. Below it, the text 'Log in to IBM' is followed by a link 'Don't have an account? Create an IBMId'. There is a form for 'IBMId' login with a 'Continue' button. Below the form, there's a checkbox for 'Remember me'. Under 'Alternative login', there's a 'Continue with Google' button. At the bottom, there's a link 'Forgot IBMId? Contact the IBMId help desk'. The right side of the page features a decorative graphic with a grid of dots and various geometric shapes like circles, diamonds, and lines in shades of blue and purple.

5. Take the tour until you reach the main page and close out of the dialog box.



6. Scroll down and select the sandbox with your name apparent; for example, **<your_name> Sandbox**.

7. Click **Automate experiments for machine learning or RAG applications**, select the Machine Learning and Model Building option.



8. Select the **AutoAI** tile.
9. Specify a name for the experiment.
10. Click **Associate a Machine Learning service instance** to this project and select the Machine Learning service instance and click reload. If you do not have Machine Learning service instance, then follow the steps on your screen to provision the service.
11. Click **Create**.
12. Use the **Browse** button to upload your CSV file from your local drive.
13. Select **No**, since this is not a time series event.
14. Select **FLAG_FOR_FRAUD_INV** as the predictable column.
 - a. Try out other features (column headings) and observe how the system recognizes Regression and Multiclassification as other potential approaches.
 - b. Before you click **Run Experiment**, discuss with your team and instructor the nuances of what lurks in the Experiment Settings tab.

Projects / Insurance Fraud / Auto Fraud Prediction

Configure AutoAI experiment
Auto Fraud Prediction

Autosaved: 1:27:34 PM

Add data sources

Drop or browse for one or more tabular data files. [Learn more.](#)

[Browse](#) or [Select from project](#)

AutoInsClaims.csv
Size: 0.28 MB | Columns: 38

Configure details

Create a time series forecast?
Enable this option to predict future activity over a specified date/time range. Data must be structured and sequential. [Learn more](#)

What do you want to predict?
Prediction columns ⓘ

Select prediction columns

DEC	LOSS_LOCATION_LAT
DEC	LOSS_LOCATION_LONG
DEC	CLAIM_AMOUNT
INT	FLAG_FOR_FRAUD_INV
STR	PRIMARY_DRIVER_ID

At this point you need to perform certain experiment settings. Before you begin in earnest, it is worth while to see some of the magic behind AutoAI.

15. Instead of FLAG_FOR_FRAUD_INV, select CLAIM_AMOUNT. Notice that the Prediction type is Regression. Now, select a column that you think would reveal a multiclassification. You are on your own here. But do not save settings nor run experiment yet; you are taking various actions to see how the system responds.
16. Once you are done with your observations on few of these columns (features), go back and select FLAG_FOR_FRAUD_INV
17. Before you run the experiment, click **Experiment settings**.
18. Include **Gradient Boosting Classifier** yet another estimator to run your experiment.

Prediction

Data source
Runtime

General Fairness Time series

Score, or optimize for those with the highest score in the shortest run time.

☐ Score only
☒ Score and run time

Algorithms to include 11 / 11

Select which of the following algorithms is to be considered when the experiment is run. The list of algorithms are based on the selected prediction type.

Search by algorithm or pipeline

Algorithm
☒ Decision Tree Classifier
☒ Extra Trees Classifier
☒ Gradient Boosting Classifier
☒ LGBM Classifier
☒ Logistic Regression
☒ Random Forest Classifier
☒ Snap Decision Tree Classifier
☒ Snap Logistic Regression
☐ Snap Random Forest Classifier

Cancel Save settings

19. Scroll further down and you will notice that by default the system will select, based on its own decision, given the myriad of algorithms to choose from, will select either, one, two, three or four estimators or algorithms. For sake of time and other credit unit hour savings, accept the default of two algorithms:

Algorithms to use 2 / 4

AutoAI will test the specified algorithms and use the top performers to create model pipelines. Choose how many top algorithms to apply. Each algorithm generates 4-5 pipelines and more algorithms increase the runtime.

1 2 3 4

20. Notice that the system is set to perform a 3-fold training and test set approach. This means that out of a possible 975 rows (records of individuals) the following events will take place:

- First, 10% of all rows (~98) will be kept away from the system at first...randomly chosen 98 rows.
- Next, the system, now has roughly 900 rows where it will break it up into three chunks (randomly grouped).
- AutoAI will build patterns from 2 of those three chunks and test itself on the third partition available to it. It may pick the top 1/3 and bottom 1/3 and test itself on the middle 1/3.
- This training set will then include two the bottom 1/3 partitions and test itself on the top 1/3 and so forth with the top and bottom 1/3, while testing itself in the middle section. The machine has the answers (supervised training) in column P (FLAG_FOR_INV).
- Once patterns are built, the system discards all training data and applied those patterns (weights) onto the hidden 10% that was kept away as test set. The metrics that we obtain from data unseen by the machine, yet obvious to human, is the telltale story behind the validation of the metrics.

Final training data set

Select what data to use when training the final pipelines. If you choose to include training data only, the generated notebooks will include a cell for retrieving the holdout data used to evaluate each pipeline.

- ☐ Training data only
- ☒ Training data and holdout data

Outlier handling

Choose whether AutoAI should exclude outlier values from the target column to improve training accuracy. If enabled, AutoAI uses the interquartile range (IQR) method to detect and exclude outliers from the final training data (training plus holdout data).

☒ Handle outliers in training and holdout data

Training and holdout method

Divide your data source into training and holdout (test) data or use a separate holdout data file.

- ☒ Split one data source
- ☐ Use separate source for holdout

Training data split ⓘ

5%  95%

Training data split: 90% — 3 — + folds | Holdout data split: 10%

- 21.
22. Move the slider up and down the line and notice the dialog box that appears as you move it above 15% test set. Bring it back to the original settings of 10%.
23. Click the **Data source** tab and start unchecking the features that you deemed unpredictable from the table above (bear in mind there are 4 pages in this selection). For this experiment, uncheck the **Claim Status** as well as all other private information (PII).
 - a. Ensure to keep ALL fields that have something to do with date. Birthdays (some folks drive with expired license) or Date_at_current_location (some folks tend to cheat on their auto insurance who tend to move from one address to another in a short time period). Keep all date-related columns.
 - b. You may decide on your own if you want to keep or remove certain attributes such as Color. Hard to argue the point that folks with Cherry red cars are up to more mischief but might be interesting to see if that may be the case.
 - c. Obviously, name, phone number, social security and other data entry ID fields are not predictable features as to why certain individuals may do fraud with their auto insurance. Furthermore, according to GDPR and the California Consumer Privacy Act such data is simply unlawful to tamper with.

Experiment settings

Prediction

Data source

Runtime

General

Time series

Join

Select features to include (32 / 38)

Select columns with data that support the prediction column.

Q Search columns

☐	Column name	Type
☐	HOUSEHOLD_ID	String
☐	DRIVER_ID	String
☐	POLICY_ID	String
☐	CLAIM_ID	String
☒	INCIDENT_CAUSE	Integer
☐	DESCRIPTION	String
☐	CLAIM_STATUS	Integer
☒	ODOMETER_AT_LOSS	Decimal
☒	LOSS_EVENT_TIME	Date
☒	CLAIM_INIT_TIME	Date

Items per page: 10 1-10 of 38 items 1 1 of 4 pages

Q Search columns

☐	Column name	Feature type
☒	POLICE_REPORT Integer	(Auto-classified) ▾
☒	CLAIMS_AT_LOSS_DATE Integer	(Auto-classified) ▾
☒	LOSS_LOCATION_LAT Decimal	(Auto-classified) ▾
☒	LOSS_LOCATION_LONG Decimal	(Auto-classified) ▾
☒	CLAIM_AMOUNT Decimal	(Auto-classified) ▾
☒	FLAG_FOR_FRAUD_INV Integer	(Auto-classified)
☐	PRIMARY_DRIVER_ID String	(Auto-classified) ▾
☒	START_DATE Date	(Auto-classified) ▾
☒	EXPIRY_DATE Date	(Auto-classified) ▾
☐	MODEL_YEAR Integer	(Auto-classified) ▾

Items per page: 10 11-20 of 38 items 2 2 of 4 pages

Search columns		
☐	Column name	Feature type
☐	MAKE String	(Auto-classified)
☐	MODEL String	(Auto-classified)
☐	PLATE String	(Auto-classified)
☒	COLOR String	(Auto-classified)
☒	INITIAL_ODOMETER Integer	(Auto-classified)
☒	LOW_MILEAGE_USE Integer	(Auto-classified)
☐	FIRST_NAME String	(Auto-classified)
☐	LAST_NAME String	(Auto-classified)
☒	GENDER String	(Auto-classified)
☒	BIRTHDATE Date	(Auto-classified)
Items per page: 10 21–30 of 38 items 3 3 of 4 pages ◀ ▶		

Search columns		
☐	Column name	Feature type
☐	SSN String	(Auto-classified)
☐	DRIVERS_LICENSE_ID String	(Auto-classified)
☒	DRIVERS_LICENSE_EXPIRY Date	(Auto-classified)
☒	DRIVERS_LICENSE_STATE String	(Auto-classified)
☒	DATE_AT_CURRENT_ADDRESS Date	(Auto-classified)
☐	CONTACT_NUMBER String	(Auto-classified)
☐	EMAIL String	(Auto-classified)
☒	COMMUTE_DISCOUNT Integer	(Auto-classified)
Items per page: 10 31–38 of 38 items 4 4 of 4 pages ◀ ▶		

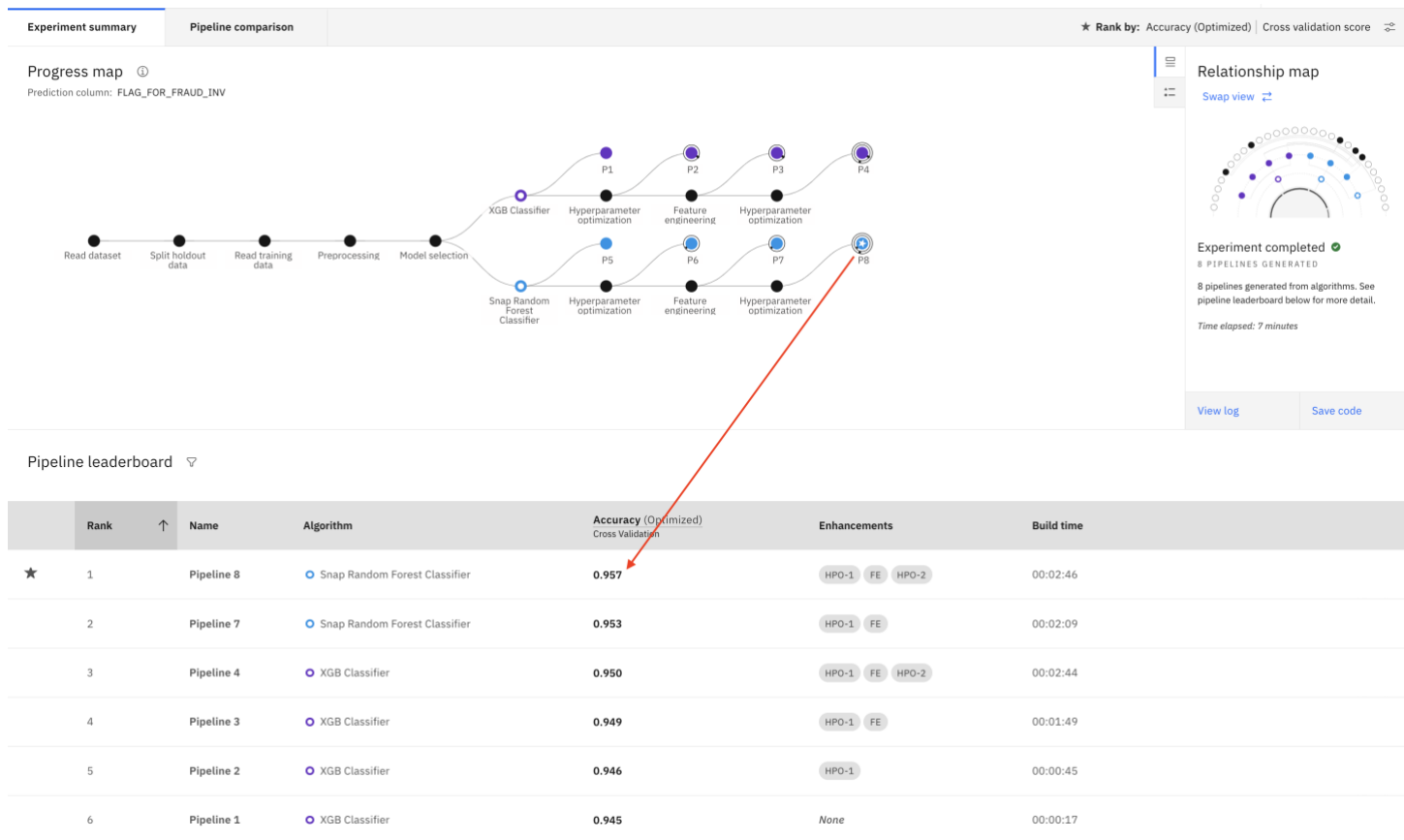
24. Save the settings.

25. Run the Experiment.

Analyze results

In this example, AutoAI experiment generated two pipelines. The duration of experiment depends completely on the size of the dataset. AutoAI selects the appropriate machine learning algorithm (in the fifth stage of the process under Model Selection) which is best suited for the dataset.

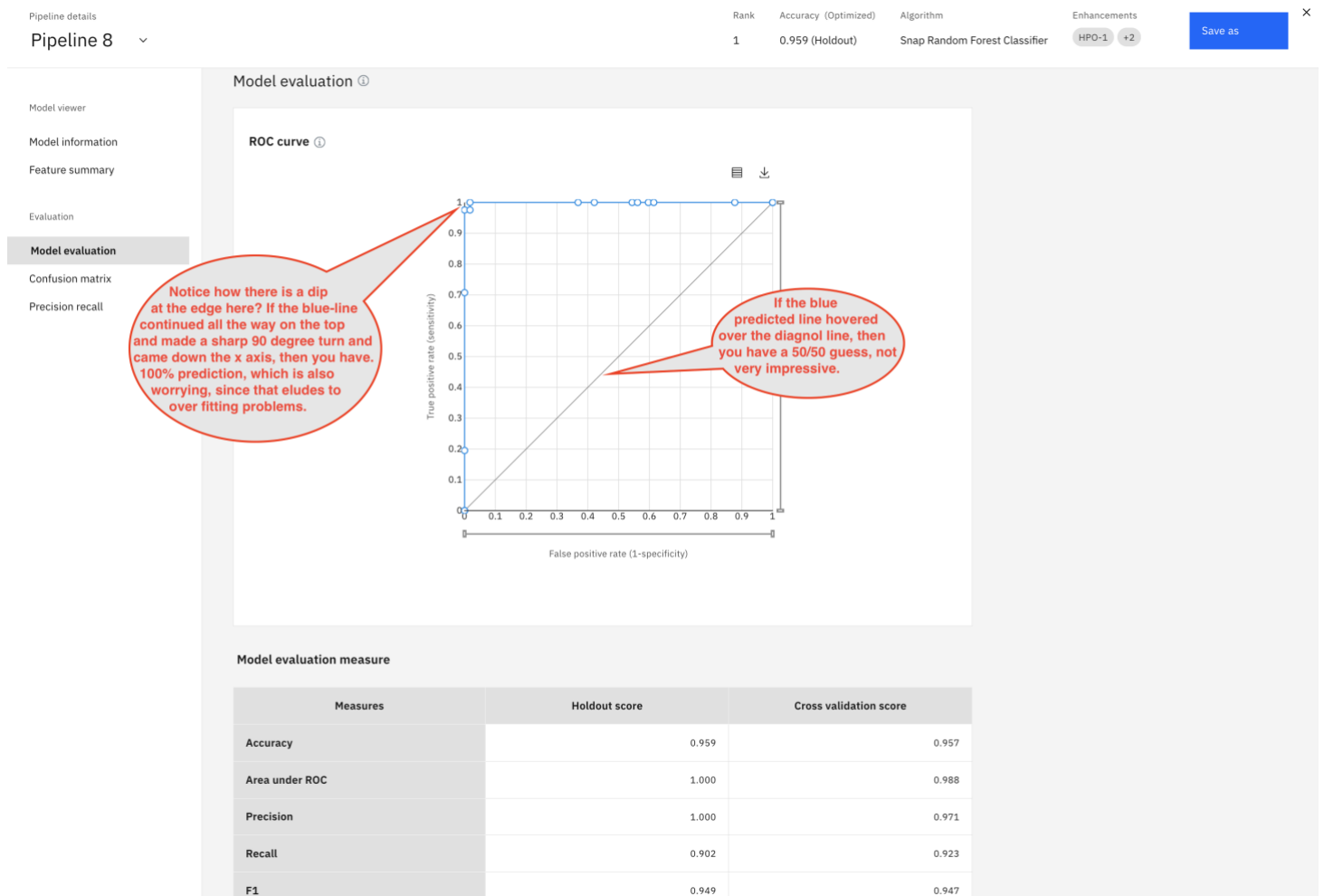
Each pipeline is run with different parameters, pipeline 2 is run on a sequence of HPO (hyper parameters optimization) & FE (feature engineering) whereas pipeline 4 includes HPO (hyper parameters optimization), FE (feature engineering) and a combination of both. All these are done on the fly! Isn't it amazing that we just have to sit and watch while AutoAI takes care of things for us and generates awesome machine learning models!! There's very minimal intervention required to get things going and in no time, we have the generated pipelines to choose from.



Note that the system 'decides' which algorithms to choose. You can always uncheck certain algorithms and force it to use fewer estimators and observe the accuracy from that machine rendered decision.

Pipeline 8, using Snap Random Forest Classifier is the winner of the two.

26. Click the highest-ranking pipeline (with the star) to see the evaluation metrics on the left side.

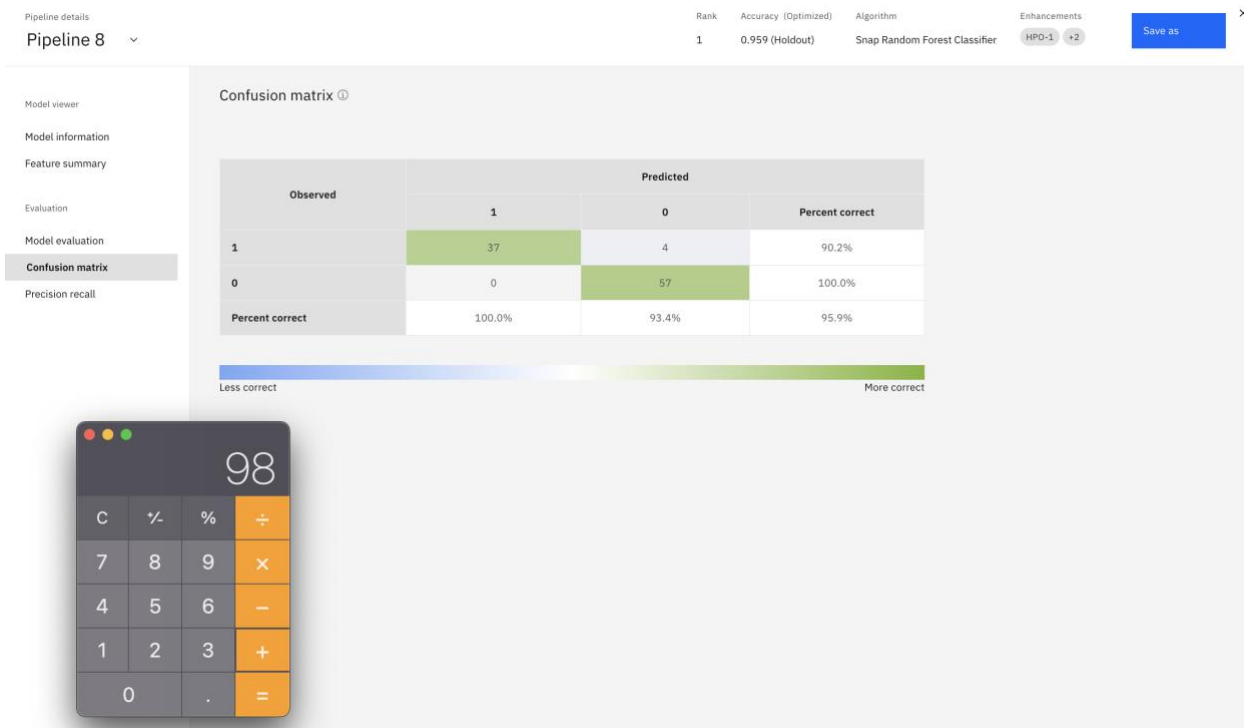


Notice that the Receiver Operating Curve, plots the True Positive against the False positive. The Receiver Operating Characteristic curve, is a graphical representation used to assess the performance of a binary classification model. It measures the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1 - Specificity) at various thresholds for a classification problem.

In a binary classification scenario, the ROC curve plots the relationship between the True Positive Rate (TPR) on the y-axis and the False Positive Rate (FPR) on the x-axis as the classification threshold is varied. The TPR represents the proportion of actual positive cases correctly predicted as positive by the model, while the FPR represents the proportion of actual negative cases incorrectly predicted as positive by the model.

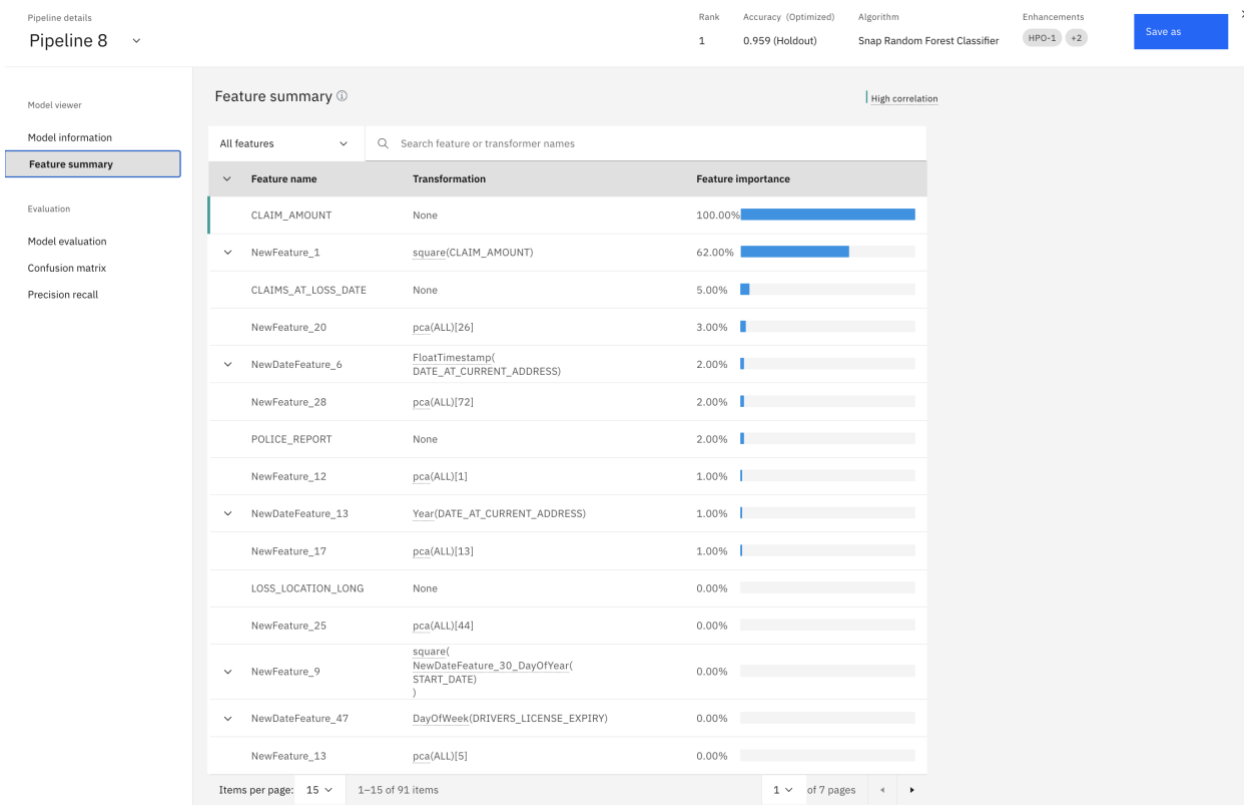
The ROC curve provides valuable insights into the model's ability to discriminate between the two classes across different threshold settings. A well-performing model will have a ROC curve that hugs the upper-left corner, indicating high TPR and low FPR across a range of threshold values. The area under the ROC curve (AUC-ROC) is a common metric used to quantify the overall performance of a classification model. A higher AUC-ROC value suggests a better discriminative ability and, in turn, a better model.

27. Click the **Confusion matrix** in the left panel.



28. The system predicted that from a total of 98 records, it predicted that 37 fraud incidents may take place, yet it missed on 4 counts. It did very good with predicting that 57 of those 98 rows will not result in fraudulent activities and it did not miss on any of those predictions. Overall, an accuracy of 96% is quite impressive.

29. Click the **Feature summary** link and it is obvious now that excessive claim amount is the number one cause of auto insurance fraud.



30. Keep this chart handy, for this is how you calculate the evaluation metrics stemming from the confusion matrix:

- **Precision** measures the ratio of true positives to total positives predicted.

$$Pr = \frac{TP}{TP + FP}$$

- **Recall** measures the ratio of positives captured out of the total true positives.

$$R = \frac{TP}{TP + FN}$$

- **F1-measure** combines precision and recall into a single metric.

$$F1 = 2 * \frac{Pr * R}{Pr + R}$$

As you have noticed we left out how to calculate the accuracy of the system. That is for you to add to your notes.

31. The last undertaking here is to save this model either as a model (pmml file) or as Jupyter Notebooks, which also can be run and used as a model for future datasets. Remember, we are not after people, merely human traits (or features) that may tell the tale of fraudulent activities.

32. Click the blue **Save as** button in the top right and select Notebook.

33. Click **Create**.

Save as

Select asset type

Model

Create a Watson Machine Learning model asset that you can test with new data, deploy to generate predictions, and trace lineage activity.

Notebook

Create a notebook if you want to view the code that created this model pipeline or interact with the model programmatically.

Define details

Name

Predicting Auto Ins Fraud - PB Notebook

Runtime environment

Runtime 22.2 on Python 3.10 AutoAI-M (4 vCPU and 16 GB RAM)

Description (optional)

Enter description here

Tags

Add tags to make assets easier to find.

Add a tag

Cancel Create

34. Click View in Project

Pipeline details

Pipeline 8

Rank 1 Accuracy (Optimized) 0.959 (Holdout) Algorithm Snap Random Forest Classifier Enhancements HPO-1

Save as

Notebook saved successfully. Predicting Auto Ins fraud - P8 Notebook was successfully saved to Auto ins fraud2. View in project

Feature summary

All features Search feature or transformer names

Feature name	Transformation	Feature importance
CLAIM_AMOUNT	None	100.00%
NewFeature_1	square(CLAIM_AMOUNT)	62.00%

High correlation

35. AutoAI, does a wonderful job in creating a meaningful Python code in Jupyter Notebooks. Take a moment and scroll thru the python code.

- Note, the first thing that you include in the first cell, is inclusion of relevant libraries and installs
- Second, is the connection credentials with the NoSQL database
- Third, is a depiction of the settings that you picked in AutoAI, the accuracy of 10%, the x-axis and the list of algorithms for the system to consider.

IBM Cloud Pak for Data Search in your workspaces Upgrade 2401692 - Armen Pischdot Dallas

Projects / Auto ins fraud2 / Predicting Auto Ins fraud - P8 No...

Package installation

Before you use the sample code in this notebook, install the following packages:

- ibm_watson_machine_learning,
- autoai-libs,
- scikit-learn,
- snapi

In []:

```
!pip install ibm-watson-machine-learning | tail -n 1
!pip install autoai-libs==1.14.13 | tail -n 1
!pip install scikit-learn==1.1 | tail -n 1
!pip install 'snapi==1.8.10' | tail -n 1
```

AutoAI experiment metadata

The following cell contains the training data connection details.
Note: The connection might contain authorization credentials, so be careful when sharing the notebook.

In []:

```
from ibm_watson_machine_learning.helpers import DataConnection
from ibm_watson_machine_learning.helpers import ContainerLocation

training_data_references = [
    DataConnection(
        data_asset_id='b0c4151a-f984-4391-8dc4-7f0860849bd9'
    ),
]

training_result_reference = DataConnection(
    location=ContainerLocation(
        path='auto_ml/6226b658-f647-4177-8d56-cbafa7419b21/wml_data/e68ddc0a-c258-48c0-8119-ca25199c895f/data/automl',
        model_location='auto_ml/6226b658-f647-4177-8d56-cbafa7419b21/wml_data/e68ddc0a-c258-48c0-8119-ca25199c895f/data/automl/model.zip',
        training_status='auto_ml/6226b658-f647-4177-8d56-cbafa7419b21/wml_data/e68ddc0a-c258-48c0-8119-ca25199c895f/training-status.json'
    )
)
```

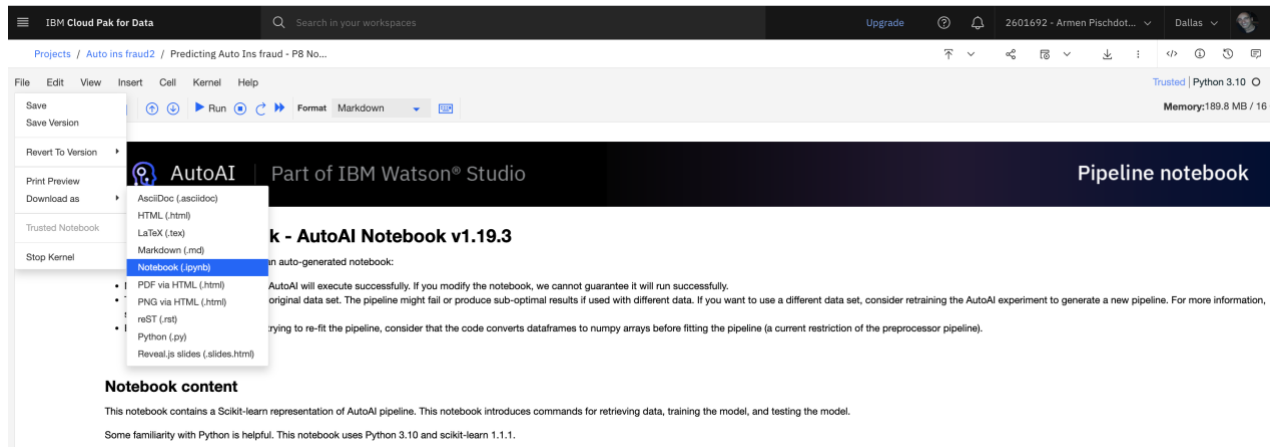
The following cell contains input parameters provided to run the AutoAI experiment in Watson Studio.

In []:

```
experiment_metadata = dict(
    prediction_type='binary',
    prediction_column='FLAG_FOR_FRAUD_INV',
    holdout_size=0.1,
    scoring='accuracy',
    csv_separator=',',
    random_state=33,
    max_number_of_estimators=2,
    training_data_references=training_data_references,
    training_result_reference=training_result_reference,
    include_only_estimators=[
        'RandomForestClassifierEstimator',
        'DecisionTreeClassifierEstimator',
        'LogisticRegressionEstimator',
        'ExtraTreesClassifierEstimator',
        'XGBClassifierEstimator',
        'LGBMClassifierEstimator',
    ],
    deployment_url='https://us-south.ml.cloud.ibm.com',
    project_id='8d4de25d-ac08-4195-b88c-92cd73e48774',
    train_sample_columns_index_list=[4, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 22, 23, 24, 27, 28, 31, 32, 33, 36],
    positive_label=1,
    drop_duplicates=True,
    include_batched_ensemble_estimators=[]
)
```

First, click the pencil icon to make it editable and then make it Trusted.

- You can now save this a local file onto your local system. From the File menu, select Download as.. and then click the **ipynb** file type.



36. We encourage you to find your own dataset and upload that instead of this CSV file. Perhaps you can predict which factors may contribute to students not completing their program; or is there an uptick of auto accidents at certain hours or certain locations?

Lab03b: Bank Lending Loan Bias

Similar to the steps above, predict if there is potential for bias in banks lending out money. Do so using this CSV file:

https://github.com/apischdo/AI_Periodic_Table/blob/main/german_credit_data_biased_training.csv